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PLACEMENT DIRECTORS

Prof. Francisco Alvarez-Cuadrado

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EDUCATION

Ph.D. in Economics, McGill University

expected May 2024

M.A. in Economics, Ewha Womans University

2018

B.A. in Economics, Ewha Womans University

2016

RESEARCH INTERESTS

Econometrics, Time-Series Analysis, Financial Economics

DISSERTATION

Inference for Models with Mixed-Frequency Data

Committee: Prof. Silvia Gonçalves (Chair), Prof. John Galbraith

JOB MARKET PAPER

“Bootstrapping Factor-MIDAS Regression Models”

Abstract: In our paper, we propose and theoretically justify bootstrap methods for the factor-augmented MIDAS (MIxed DAta Sampling) regression models, where the factors are estimated from a large panel dataset. We derive the asymptotic distribution of the models under the assumption that $\sqrt{T}/N \rightarrow c$, $0 \leq c < \infty$, where T is the time series dimension and N is the cross-sectional dimension. When $c \neq 0$, we find that there exists an asymptotic bias due to the two-step estimation procedure. The asymptotic bias depends not only on the cross-sectional dependence but also on the serial dependence of the idiosyncratic term in the factor model. We propose a general residual-based bootstrap and provide high-level conditions under which the bootstrap distribution of an estimator is consistent. We verify these conditions for a bootstrap procedure, autoregressive sieve with cross-sectional dependent bootstrap. This bootstrap method is robust to cross-sectional dependence and serial dependence. We show its finite sample performance in Monte Carlo simulation and find that the proposed bootstrap procedure outperforms the benchmark bootstrap and plug in bias estimator. Furthermore, we apply the bootstrap method to nowcast US GDP growth rate. We show that the bootstrap method recovers the distortion of the confidence interval due to the bias.

WORKING PAPERS

“Bootstrap inference for group factor models”, with Sílvia Gonçalves and Benoit Perron, work in progress

Abstract: Recently, Andreou et al. (2019) have proposed a test for common factors based on canonical correlations between factors estimated separately from each group. We propose and theoretically justify a simple bootstrap test that avoids the need to estimate the bias and variance of the canonical correlations explicitly. We verify these conditions for a wild bootstrap scheme similar to the one proposed in Gonçalves and Perron (2014). Simulation experiments show that this bootstrap approach leads to rejection rates closer to the nominal level in all of our designs compared to the asymptotic framework.

TEACHING EXPERIENCE

McGill University - Lecturer

Mathematics for Economists (graduate)	Summer 2023
Economic Statistics (undergraduate)	Spring 2023
Mathematics for Economists (graduate)	Summer 2022
Econometrics 2 Honours (undergraduate)	Winter 2022

McGill University - Teaching Assistant

Macroeconomic Analysis & Application (undergraduate)	Spring 2023
Econometrics 1 Honours (undergraduate)	Fall 2022
Microeconomic Theory 1 (graduate)	Fall 2021
Macroeconomic Theory 1 (graduate)	Fall 2021
Applied Cross-Sectional Methods (graduate)	Spring 2021
Applied Time-Series & Forecast (graduate)	Spring 2021
Economic Statistics (undergraduate)	Fall 2018 - Fall 2020

Ewha Womans University - Teaching Assistant

Econometrics (graduate)	Fall 2017
Money and Banking (undergraduate)	Fall 2017
Financial Econometrics (undergraduate)	Fall 2017
Introduction to Financial Engineering (undergraduate)	Fall 2017
Financial Economics (undergraduate)	Spring 2017
Microeconometrics (graduate)	Spring 2016
Applied Econometrics (undergraduate)	Spring 2016 - Spring 2017
Statistics for Economics (undergraduate)	Spring 2016 - Fall 2017

CONFERENCE AND SEMINAR PARTICIPATION

2023	Canadian Econometrics Study Group , October (Scheduled)
	International Association of Applied Econometrics (coauthor presented), June
	Econometric Society North American Summer Meeting, June
	Society of Financial Econometrics, June
	Canadian Economic Association (coauthor presented), June
	CIREQ Econometrics Conference, May
	Virtual Workshop for Junior Researchers in Time Series, April
	CIREQ, McGill University, April
2022	International Symposium on Econometric Theory and Applications, July
	Canadian Economic Association, June
	CIREQ, McGill University, April

HONORS AND SCHOLARSHIPS

2018 - 2024 Graduate Excellence Award, McGill University

OTHER RESEARCH EXPERIENCE

Prof. Silvia Gonçalves, McGill University
Prof. Hosin Song, Ewha Womans University
Prof. Vladimir Hlasny, Ewha Womans University

Spring 2022
Spring - Winter 2017
Fall 2016 - Spring 2017

TECHNICAL STRENGTHS

Matlab, STATA, R Python

LANGUAGE

English (Fluent), Korean (Native), French (Intermediate)

REFERENCES

Prof. Silvia Gonçalves, McGill University

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Prof. John Galbraith, McGill University

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Prof. Benoit Perron, Université de Montréal

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